Prefix orbits from a singleton end

Human Compression project: L044 continuation

Let $G = \langle \rho_0, \dots, \rho_{r-1} \rangle \le \mathrm{Alt}(\Omega)$ be transitive with a connected string C-marking, and suppose every maximal omission is intransitive. Suppose G_0 has the singleton orbit $\{a\}$. Put $P_k = G_{\le k}$ and $C_k = a^{P_k}$. The other G_0-orbit is understood to be its entire complement, as in the complete singleton-end interface.

Lemma 1 (Strict growth without an alternating kernel)**.** *One has $|C_0| = 2$ and $C_{k-1} \subsetneq C_k$ for $1 \le k < r$. Every mark of index at least $k+2$ fixes C_k pointwise. In particular $|C_k| \ge k+2$.*

Proof. Every mark except ρ_0 fixes a. Transitivity makes $a^{\rho_0} \ne a$, proving the first assertion. A mark of index at least $k+2$ centralizes P_k and fixes a, so fixes its entire P_k-orbit. If $C_k = C_{k-1}$, the marks up to index k preserve this set and all later marks fix it pointwise by the same observation applied to P_{k-1}. It would be G-invariant and hence equal to Ω. But $P_{k-1} \le G_k$, so this would make that omission transitive, a contradiction. The size bound follows. No claim of a pure alternating kernel is used or implied. □

Lemma 2 (The actual sides of a later cut)**.** *If a colour $j \ge 2$ has exactly one ambient-eligible edge, its two omission orbits are $C = C_{j-1}$ and $A = \Omega \setminus C$. The suffix $G_{>j}$ fixes C pointwise. Every shared label at this cut is below j.*

Proof. The suffix centralizes P_{j-1} and fixes a, so fixes C. Thus C is exactly the G_j-orbit of a. A single crossing edge can connect all omission orbits only if there are exactly two. The suffix cannot act on both, giving the final assertion. □

For the next theorem retain the complete acquired support, small-tail and critical-dihedral proofs, including their explicit smaller-degree alternating induction. Assume $n = |\Omega| \ge 12$ and the contrary range $r > \lfloor (n-1)/2 \rfloor$. These are precisely the hypotheses under which the complete critical singleton reduction gives $n = 2r \ge 14$ and $G_0 = \mathrm{Alt}(\Omega \setminus \{a\})$.

Theorem 3 (The first shared indices at a singleton end)**.** *Suppose a later unique cut $j \ge 2$ has a first shared label g. Then $g \ge 3$ and*

$$|\operatorname{supp}(P_{g-1})| \le 2g + 1. \tag{1}$$

The small-index exclusions use the actual singleton point and the complete critical-dihedral-tail proof, without assuming that the cut adjacent to a supported head is itself unique.

Proof. Reverse the entire string. Its crossing is $i = r-1-j$; the last shared label is $h = r-1-g$. If $i > 0$, the group preceding this crossing is $G_{>j}$ and fixes C pointwise by Lemma 2. Both omission orbits are nontrivial: C has at least three points, and a shared label acts on A. Thus the complete interior support alternative applies, using its transitive or intransitive case according to the actual action of $G_{>j}$ on A. It excludes $h = r-1$ in the contrary range. If $i = 0$, the complete endpoint shared-last-label theorem gives the same exclusion. Consequently $g > 0$.

All labels below g are unshared. They act on the same omission orbit, since a change of sides among consecutive nonidentity marks would disconnect the noncommutation diagram. This side is C, because ρ_0 moves a. The entire reversed tail is therefore exactly the actual supported prefix P_{g-1}. The complete support bound, interior or endpoint as appropriate, gives

$$n - |\operatorname{supp}(P_{g-1})| \geq 2h + 1.$$

Substituting $n = 2r$ and $h = r - 1 - g$ proves (1). For $g = 1$ this bounds the support of the nonidentity even involution ρ_0 by three, impossible. Hence $g \geq 2$.

Suppose $g = 2$. The support of P_1 has at most five points, and its orbit C_1 has at least three by Lemma 1. If P_1 is intransitive on its support, there is exactly one other nontrivial orbit E, with $|C_1| = 3$ and $|E| = 2$. Write $C_1 = \{a, b, c\}$, where $b = a^{\rho_0}$. The two restrictions to C_1 are $(a\ b)$ and $(b\ c)$: ρ_1 fixes a, and the displayed orbit is transitive. Global evenness then forces both marks to exchange the pair $E = \{u, v\}$. Both fix every other point.

The mark ρ_2 fixes a and centralizes $\rho_0 = (a\ b)(u\ v)$, so fixes b and preserves E. Every mark of index at least three centralizes P_1. Its two nontrivial orbits have distinct sizes three and two, so such a mark preserves E as well. Hence all of G preserves this proper pair, contrary to transitivity. Thus P_1 is transitive on its support.

A connected even rank-two marking cannot act transitively on at most four points: degrees below four have no nonidentity even involutions, and the involutions of A_4 commute. Its support therefore has exactly five points. The two generators are double transpositions. Their product has order two, three or five by its cycle type in A_5; orders two and three give groups of orders at most four and six, neither transitive on five points. Its order is therefore five, and the actual group is the natural D_{10}. Its two distinct reflection generators have the standard ordered marking after a change of vertex names: in the affine action on $\mathbb{F}_5$, a translation and nonzero scaling normalize the two reflection parameters. Apply now the complete critical-dihedral-tail lemma at the original cut, in the already established reversed orientation. Its supported tail past $h = r - 3$ is exactly this actual P_1, of degree five, and its fixed deficit is $n-5 = 2r-5 = 2h+1$. If the reversed crossing is interior, its preceding subgroup fixes the other orbit C pointwise by Lemma 2; the endpoint version has no such preceding subgroup. These are precisely the two cases of the retained lemma, which excludes both for $n \geq 12$. Thus $g = 2$ is impossible as well. No claim that the separate original colour 2 has a unique eligible edge, or that its opposite suffix is transitive, has been inserted. □

Attribution and boundary. This develops the remaining singleton route in the source Propositions 5.28–5.32. The complete reversed support and critical-dihedral proofs are retained with their attribution to Cameron–Fernandes–Leemans–Mixer. The strict prefix growth uses the distinguished point fixed by every later mark; its argument resembles the acquired head-prefix growth but does not import that theorem's alternating kernel. The exterior pair exclusion above is a literal permutation argument. First shared indices $g \geq 3$, unshared later cuts, and the complete singleton and principal source assembly remain. No whole-original, question, numerical or formal credit is claimed.